



Customer Behaviour Analysis

A structured retail/e-commerce dataset used for customer behavior analysis and business decision-making 4 exploring purchase patterns, segmentation and revenue optimization through Python EDA, SQL and Power BI.

About the Dataset



Data Mix:

- **Numerical:** Age, Purchase Amount, Previous Purchases
- **Categorical:** Category, Payment Method, Season

Used for **customer behavior analysis** + **business decision-making**

Type

Retail / E-commerce

Level

Customer Transaction Level

Nature

Structured (Tabular)

Why This Data Matters



High-Value Customers

Identify frequent buyers and one-time purchasers



Purchase Patterns

Understand what, how often, and how much customers buy



Discount Strategy

Optimize promo codes and campaign effectiveness



Customer Lifetime Value

Increase CLV through targeted retention efforts

Business Problems to Solve

Low Customer Retention

Who are frequent buyers vs. one-time buyers?

Ineffective Marketing Campaigns

Do discounts increase purchases? Which customers respond to promo codes?

Product Demand Understanding

Which category performs best? Which products generate the highest revenue?

Payment Behavior Insights

Which payment methods are preferred? Are digital payments increasing?

Seasonal Demand Variation

Does season impact sales? Which season drives maximum purchases?

Customer Segmentation

Identify high-value, frequent, and discount-sensitive customers

Data Dictionary

Customer ID	Unique identifier for each customer
Age / Gender	Customer demographics
Item Purchased / Category	Product & category (Clothing, Electronics, etc.)
Purchase Amount	Amount spent by the customer
Location	Customer's geographic location
Size / Color	Product size (S, M, L) & color
Season	Season of purchase (Winter, Summer, etc.)
Review Rating	Customer rating for the product
Subscription Status	Whether customer is subscribed (Yes/No)
Shipping Type	Type of shipping selected
Discount Applied	Whether discount was applied
Previous Purchases	Number of past purchases
Payment Method	Mode of payment
Frequency of Purchases	Purchase frequency (Weekly, Monthly, etc.)

EDA — Exploratory Data Analysis (Python)



Load Data

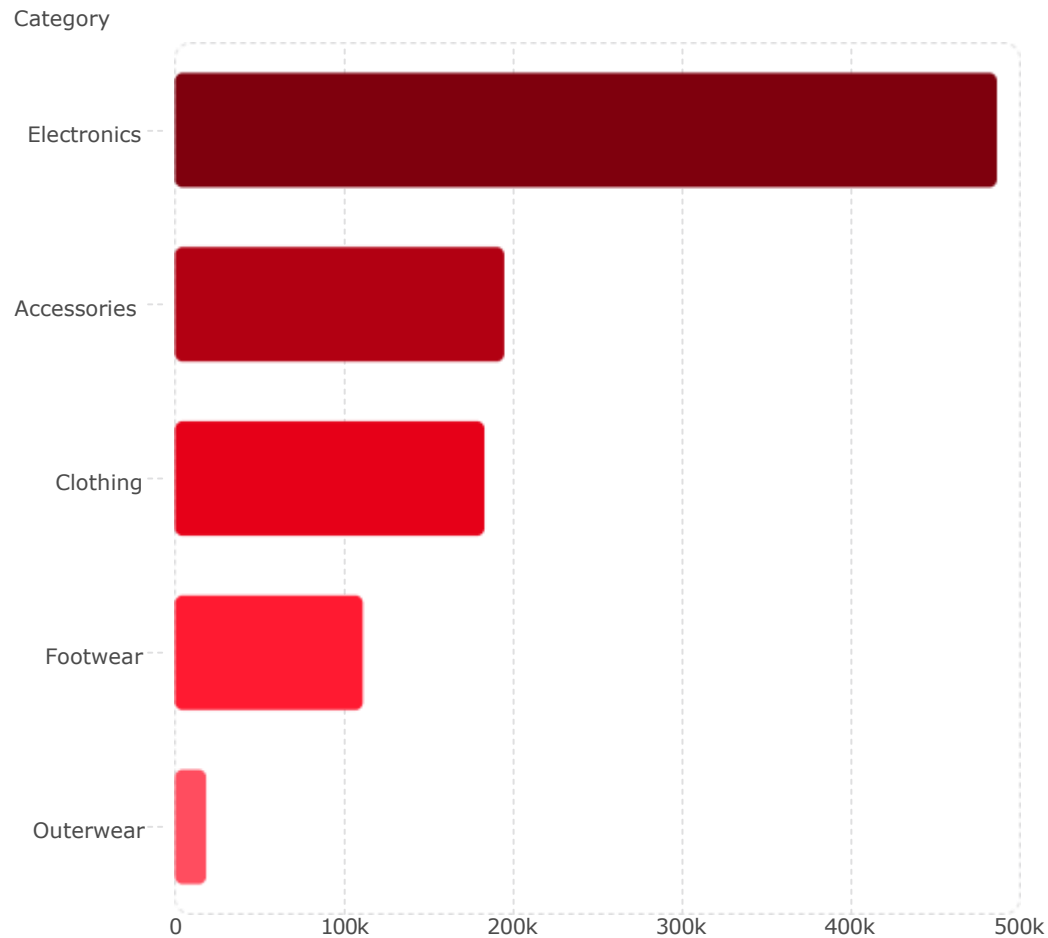
Initial Exploration

Data Cleaning

Export to SQL

Key cleaning steps included: category mapping corrections, smart imputation (product-level mean for ratings, mode for clothing sizes, zero for new customers), duplicate removal, and column standardization for SQL/Power BI compatibility.

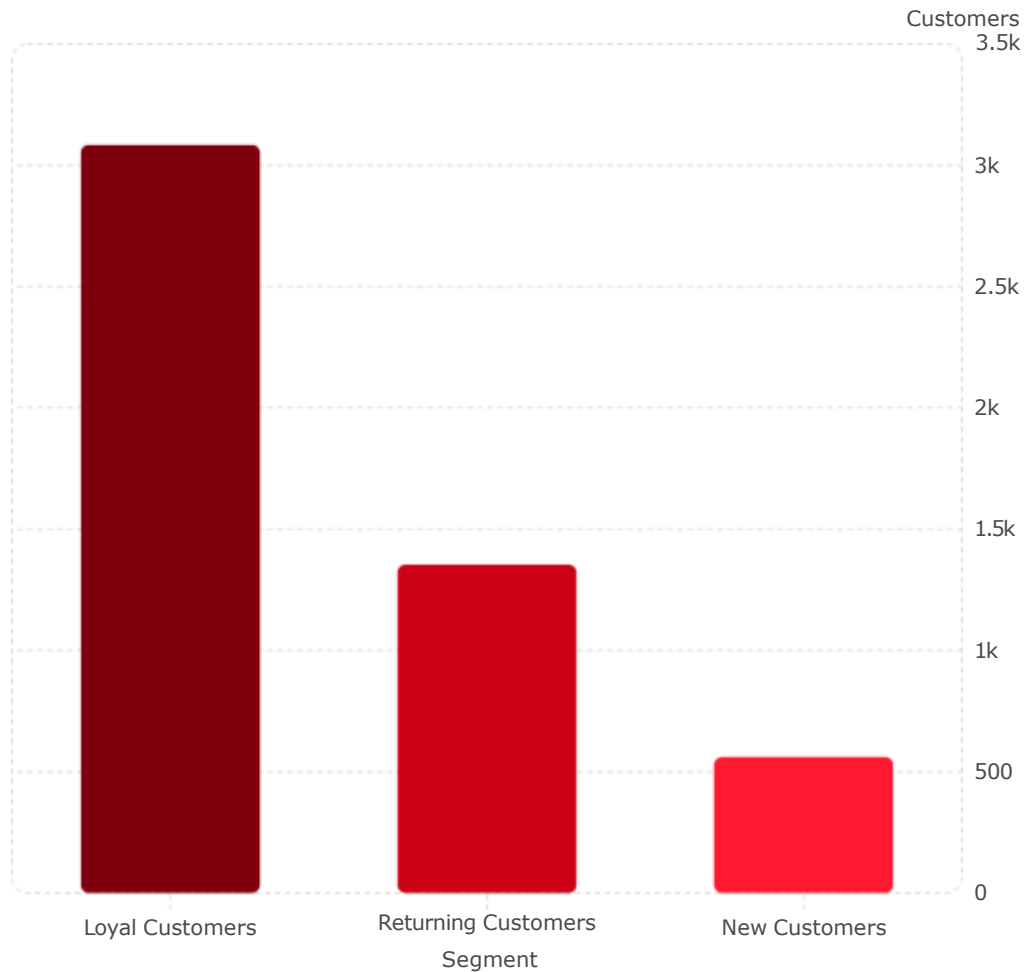
SQL Analysis — Revenue & Discounts



Key Findings

- **Electronics** leads revenue at \$486K
- Discounts **increase avg. purchase**: \$219 (discounted) vs. \$182 (no discount)
- Subscribers spend **63% more**: \$270.58 avg vs. \$165.72 for non-subscribers
- Male: \$391K · Female: \$352K · Other: \$251K revenue

Customer Segmentation & Age Insights



Segmentation Logic

Customers segmented by previous purchases into **Loyal** (3,085), **Returning** (1,354), and **New** (561).

Revenue by Age Group

- ♦ **51+:** \$399K (40.2%)
- ♦ **36-50:** \$267K (26.8%)
- ♦ **26-35:** \$177K (17.8%)
- ♦ **18-25:** \$151K (15.2%)

Repeat buyers (5+ purchases) show strong subscription uptake: **1,190** subscribed vs. 2,763 not subscribed.

Power BI Dashboard — Key Metrics

5000

Total Customers

30

Unique Items

3.62

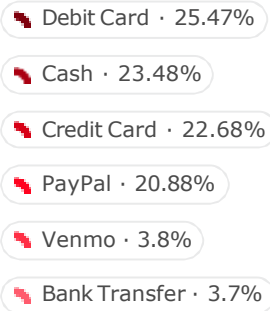
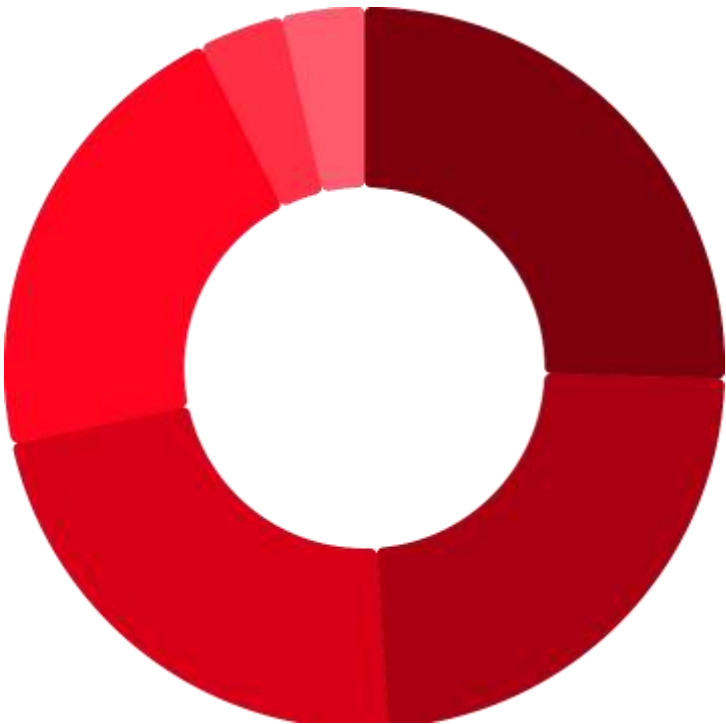
Average Rating

994K

Total Spend (\$)

198

Avg. Spend (\$)



Highlights

- **Revenue by Gender:** Male \$391K (39.3%), Female \$354K (35.4%), Other \$251K (25.3%)
- **Top Locations:** New York \$144K, Los Angeles \$126K, Phoenix \$119K
- **Seasonal:** Spring \$244K leads; Fall lowest at \$68K
- **Shipping:** Standard (1,199 orders) vs. Express (avg. \$358/order)

Key Takeaways



Electronics Dominate

Highest revenue category at \$486K
— focus inventory and marketing
here



Discounts Raise AOV

Discounted orders average \$219 vs.
\$182 — target strategically



Loyal Customers Lead

3,085 loyal customers drive the
majority of repeat revenue



51+ Age Group Spends Most

\$399K revenue — highest-spending demographic by far



Subscribers Spend 63% More

\$270 avg vs. \$166 — subscription programs boost lifetime
value